

2.1 #s: 1, 3, 5, 8

2.2 #s: 6, 8, 9, 10

Homework 3, 2.1 & 2.2

by Tyler Trotter

1: a) If $|x-5| < 1$, then x is a number greater than 4 and less than 6;

If true, then if $x \geq 5$, $|x-5| = x-5 < 1$ by def. of Abs. Val. $\Rightarrow x < 6$.

If true, then if $x < 5$, $|x-5| = -(x-5) < 1$ by Abs. Val. $\Rightarrow -x+5 < 1 \Rightarrow -x < -4 \Rightarrow x > 4$.

b) If $|x-3| < 1/2$ and $|y-3| < 1/2$, then $|x-y| < 1$.

$$-1/2 < x-3 < 1/2 \text{ and } -1/2 < y-3 < 1/2.$$

$$-1/2+3 < x < 1/2+3 \text{ and } -1/2+3 < y < 1/2+3$$

$$2.5 < x < 3.5 \text{ and } 2.5 < y < 3.5$$

$$2.5-3.5 < x-y < 3.5-2.5 \Rightarrow -1 < x-y < 1 \Rightarrow |x-y| < 1$$

c) If $|x-a| < 1/2$ and $|y-b| < 1/2$, then $|x+y-(a+b)| < 1$.

If true, then:

$$-1/2 < x-a < 1/2 \text{ and } -1/2 < y-b < 1/2;$$

$$-1/2+(-1/2) < x-a+y-b < 1/2+1/2$$

$$-1 < x+y-a-b < 1$$

$$-1 < x+y-(a+b) < 1$$

$$|x+y-(a+b)| < 1$$

3: a) $\{a_n\} = 1, 3, 5, 7, 9, \dots, \boxed{2n-1}, \dots$

b) $\{a_n\} = 1, -1/2, 1/4, \dots, \boxed{\left(-\frac{1}{2}\right)^{n-1}}, \dots$

c) $\{a_n\} = 1, \frac{1}{2}, \frac{1}{6}, \frac{1}{24}, \dots, \boxed{\frac{1}{n!}}, \dots$

5. $\lim_{n \rightarrow \infty} \frac{2n-1}{3n+1} \rightarrow \frac{2}{3}$.

Proof: Given an $\epsilon > 0$, we must show $\exists N$ s.t. $n > N \rightarrow \left| \frac{2n-1}{3n+1} - \frac{2}{3} \right| < \epsilon$.

$$\left| \frac{6n-3-6n+2}{3(3n+1)} \right| < \epsilon \Rightarrow \frac{1}{9n+3} < \epsilon; \frac{1}{9\epsilon} - \frac{1}{3} < n.$$

So, when $N = \frac{1}{9\epsilon} - \frac{1}{3}$, then $n > N$ implies $\left| \frac{2n-1}{3n+1} - \frac{2}{3} \right| < \epsilon$.

21 Hs: 8

22 Hs: 3, 5, 9, 10

Homework 3, cont.

8: $\lim \{ \sqrt{n+1} - \sqrt{n} \} \rightarrow 0$

Proof: Given $\varepsilon > 0$, we seek to show $\exists N$ s.t.:

$$n > N \rightarrow |\sqrt{n+1} - \sqrt{n}| < \varepsilon.$$

$$\text{So, } |\sqrt{n+1} - \sqrt{n}| = \sqrt{n+1} - \sqrt{n}$$
$$(\sqrt{n+1} - \sqrt{n})^2 < \varepsilon^2$$

$$n+1 - 2\sqrt{n+1} \cdot \sqrt{n} - n < \varepsilon^2$$

$$1 - 2\sqrt{n} \sqrt{n+1} = 1 - 2\sqrt{n^2+n} < \varepsilon^2$$

$$\text{So, } \sqrt{n^2+n} > \frac{1-\varepsilon^2}{2}$$

$$n^2+n > \left(\frac{1-\varepsilon^2}{2} \right)^2$$

$$\left(\frac{1-\varepsilon^2}{2} \right)^2 < n^2+n \leq n^2+n^2 = 2n^2,$$

$$n^2 > \frac{\left(\frac{1-\varepsilon^2}{2} \right)^2}{2}, \text{ then } n > \sqrt{\frac{\left(\frac{1-\varepsilon^2}{2} \right)^2}{2}}, \text{ so } \frac{1-\varepsilon^2}{2\sqrt{2}} < n$$

$$\text{So, } N = \frac{1-\varepsilon^2}{2\sqrt{2}}, \text{ then } n > N \text{ implies}$$

$$|\sqrt{n+1} - \sqrt{n}| < \varepsilon.$$

2.2 #3, 5, 9, 10

Homework 3, cont.

3: $\lim \frac{1}{\sqrt{n}} \rightarrow 0$

work: $\left| \frac{1}{\sqrt{n}} - 0 \right| < \varepsilon$; $\left| \frac{1}{\sqrt{n}} \right| < \varepsilon$; $\frac{1}{\sqrt{n}} < \varepsilon$; $\frac{1}{n} < \varepsilon^2$

$n > \frac{1}{\varepsilon^2}$. So $N = \frac{1}{\varepsilon^2}$

proof: The sequence $\frac{1}{\sqrt{n}}$ converges to 0 if for all $\varepsilon > 0$, there exists a number $N = \frac{1}{\varepsilon^2}$ such that:

$$\left| \frac{1}{\sqrt{n}} - 0 \right| < \varepsilon$$

$$\left| \frac{1}{\sqrt{n}} \right| = \frac{1}{\sqrt{n}} < \varepsilon$$

for all $n > N = \frac{1}{\varepsilon^2}$, $\left| \frac{1}{\sqrt{n}} - 0 \right| < \varepsilon$.

5: $\lim (\sqrt{n^2 + n} - n) \rightarrow 1/2$

Proof: Let $\varepsilon > 0$.

$$\left| a_n - \frac{1}{2} \right| = \left| \sqrt{n^2 + n} - n - \frac{1}{2} \right| = \frac{1}{2} \left| 2\sqrt{n^2 + n} - 2n - 1 \right|$$

$$= \frac{1}{2} \left| \frac{(2\sqrt{n^2 + n} - (2n + 1))(2\sqrt{n^2 + n} + 2n + 1)}{2\sqrt{n^2 + n} + 2n + 1} \right|$$

$$= \frac{1}{2} \left| \frac{4(n^2 + n) - (2n + 1)^2}{2\sqrt{n^2 + n} + 2n + 1} \right| = \frac{1}{2} \left| \frac{4n^2 + 4n - 4n^2 - 4n - 1}{2\sqrt{n^2 + n} + 2n + 1} \right|$$

$$= \frac{1}{2} \left(\frac{1}{2\sqrt{n^2 + n} + 2n + 1} \right) < \frac{1}{n} < \varepsilon$$

set $\frac{1}{N} = \varepsilon$ or $N = \frac{1}{\varepsilon}$. Hence, for all $n > N$

$$\left| a_n - \frac{1}{2} \right| < \varepsilon.$$

#s: 9, 10

Homework 3, cont.

9: Does the sequence $\{\cos(n\pi/3)\}$ have a limit?

$$\cos(\pi/3) = \frac{1}{2}, \cos(2\pi/3) = -\frac{1}{2}, \cos(\pi) = -1$$

$$\cos(4\pi/3) = -\frac{1}{2}, \cos(5\pi/3) = \frac{1}{2}, \cos(2\pi) = 1$$

For any $n \in \mathbb{N}$, $n \pmod{3} \in \{0, 1, 2\}$ will result in the above forms and since there is no bounding or unbounding multiplier, $\cos$ remains cyclic taking any of the above form for any n . Thus, no convergence takes place.

10: Give an example of a sequence $\{a_n\}$ which does not converge but for which the sequence $\{|a_n|\}$ does converge.

$$-n^{1/2}$$

$$\boxed{\begin{array}{l} a_n = -n^{1/2} \\ |a_n| = |-n^{1/2}| \end{array}}$$